

Vision-Based Autonomous Pursuit-Evasion in Urban Environments with Recurrent Policies

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Problem Overview

Objective

Develop an autonomous evader vehicle capable of escaping a pursuing agent in a partially observable urban environment using Reinforcement Learning.

Main Concepts

- Webots simulation environment
- PPO-based autonomous navigation
- Partial observability
- Urban pursuit-evasion scenario
- Vision-based decision making

Scenario

- Evader uses only onboard sensors
- Pursuer receives periodic GPS updates
- Dynamic and asymmetric information environment

System Architecture

Environment

- Webots “City Traffic” world
- Urban roads and buildings
- Static obstacles



Evader

- Camera perception
- LiDAR obstacle detection
- PPO policy
- Steering and throttle control

Pursuer

- Reactive pursuit strategy
- Live GPS-based tracking

Method and Reward Design

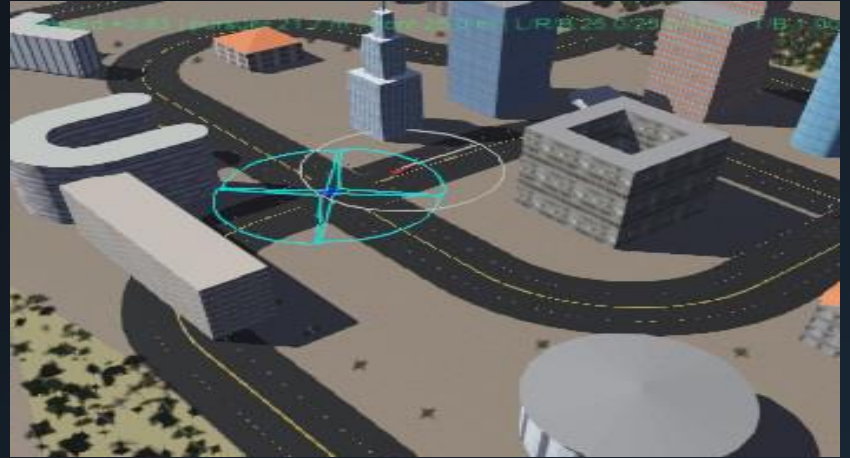
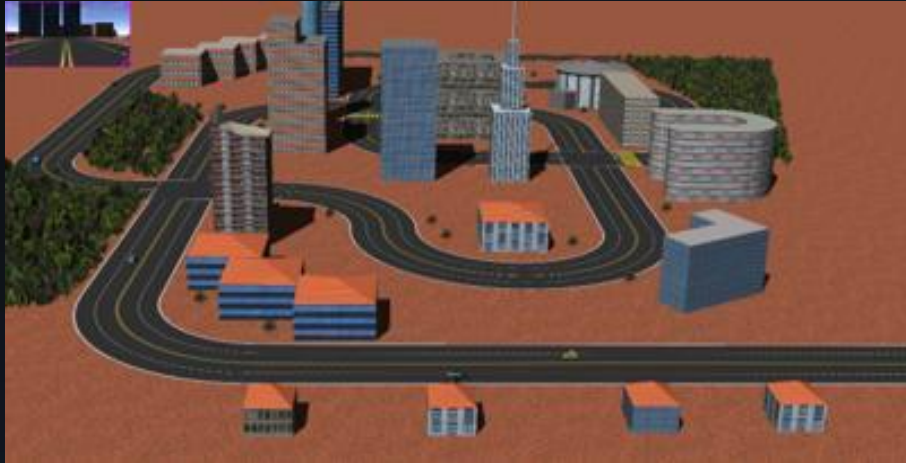
PPO + LSTM Policy

- PPO handles stable policy optimization
- LSTM adds temporal memory for better decisions over time

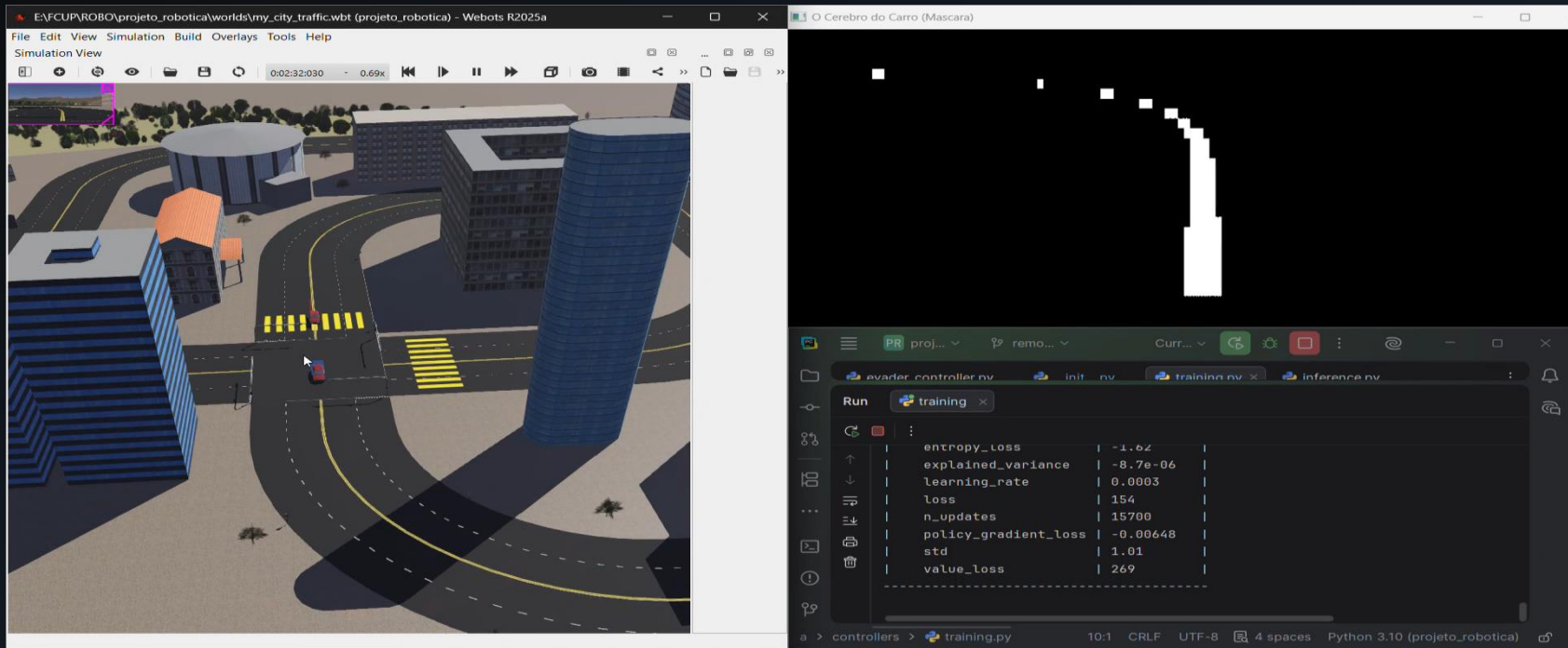
Reward Function

- Positive reward for surviving longer in the environment
- Positive reward for keeping the evader moving
- Positive reward for moving away from nearby obstacles
- Penalties are used for unsafe behavior, such as collisions
- The reward encourages active evasion instead of staying still

Initial Demo



Trying out a different approach



Challenges and Next Steps

- Improve reward balance to avoid unwanted behaviors
- Train the PPO+LSTM policy for longer and compare checkpoints
- Refine the pursuer behavior to make the evasion task more realistic
- Improve obstacle avoidance using LiDAR-based feedback
- Add or improve vision/line-of-sight detection
- Evaluate performance with survival time, collision rate, and escape rate